Differential Equations and Slope Fields

Verifying solutions to differential equations

1 Verify that $y = Ce^{3x}$ is a solution to the differential equation $\frac{dy}{dx} = 3y$ for any constant C.

2 Verify that $y = x^2 + C$ is a general solution to $\frac{dy}{dx} = 2x$, and find the particular solution satisfying $y(1) = 5$.

Solving separable differential equations

3 Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 3$.

4 Solve $\frac{dy}{dx} = 2xy$ given $y(0) = 1$.

5 Solve $\frac{dy}{dx} = \frac{\cos x}{y^2}$ given $y(0) = 2$.

Slope fields

6 For the differential equation $\frac{dy}{dx} = x - y$, find the slope at the points $(0, 0)$, $(1, 0)$, and $(1, 1)$, as would be drawn on a slope field.

7 Explain how a slope field for $\frac{dy}{dx} = ky$ (exponential growth/decay) would look different for $k > 0$ versus $k < 0$.

Exponential growth and decay models

8 A bacteria culture grows according to $\frac{dP}{dt} = 0.2P$, with $P(0) = 500$. Find $P(t)$ and the population after 10 hours.

9 A radioactive substance decays according to $\frac{dA}{dt} = -0.05A$. If $A(0) = 100$ grams, find how long it takes to decay to 50 grams (the half-life).

Matching slope fields to differential equations

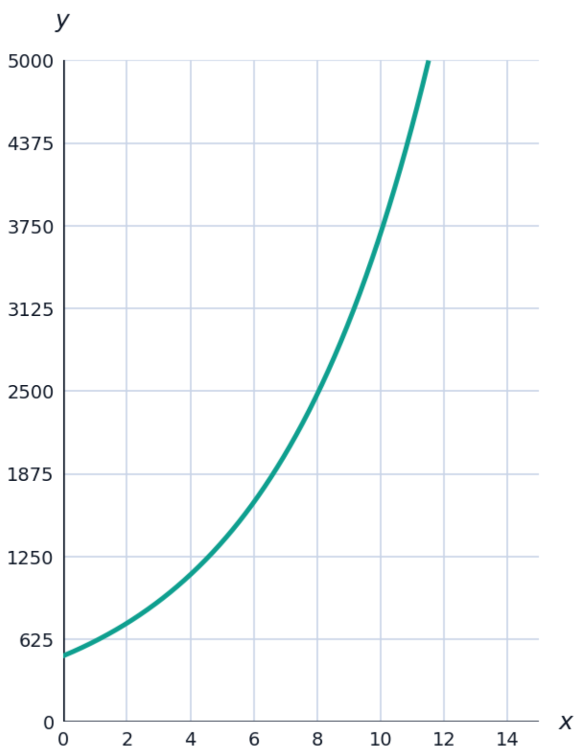
10 A slope field shows horizontal segments (slope 0) along the entire x-axis and along the entire y-axis, with slopes elsewhere resembling the product xy. Identify the differential equation this slope field represents.

Newton's Law of Cooling

11 An object cools according to $\frac{dT}{dt} = -k(T-70)$, where $70°$F is room temperature. If $T(0) = 200°$F and $k = 0.1$, find $T(t)$.

Visualizing an exponential growth solution

12 The graph shows the bacteria population $P(t) = 500e^{0.2t}$ from the earlier problem. Use the graph to estimate how long it takes for the population to double from its initial value of 500, then verify algebraically using $\ln 2$.



A logistic-style differential equation

13 A population satisfies $\frac{dP}{dt} = 0.1P\left(1 - \frac{P}{200}\right)$. Find the equilibrium solutions (where $\frac{dP}{dt} = 0$), and explain which one is stable for a population starting near $P = 10$.

Additional separable equations

14 Solve $\frac{dy}{dx} = \frac{y}{x}$ given $y(1) = 6$.

15 Solve $\frac{dy}{dx} = \frac{1}{y}$ given $y(0) = 4$.

A mixing problem modeled by a differential equation

16 A tank contains 50 L of pure water. Brine with concentration 3 kg/L flows in at 2 L/min, and the well-mixed solution drains at the same rate. If $y(t)$ is the amount of salt (kg) at time t, the differential equation is $\frac{dy}{dt} = 6 - \frac{2y}{50} = 6 - \frac{y}{25}$. Find the equilibrium value of y, and explain what it represents physically.